

A NUMERICAL METHOD FOR SOLVING THE EIKONAL EQUATION ON SIMPLICIAL TESSELLATIONS*

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Abstract. In this paper, we describe a method for solving the Eikonal equation on domains in $\mathbb{R}^2$ or $\mathbb{R}^3$ tessellated by triangles or tetrahedra. Our method generalizes to simplices in higher dimensions, owing to our choice of solution representation under the Jacobi family of orthogonal polynomials on the simplex. First, we describe a spectral decomposition for functions supported on the simplex in terms of their weighted L_2 expansion coefficients under the Jacobi basis. Computing this expansion requires an accurate and efficient quadrature, and we provide a novel optimization routine based on approximate joint-diagonalization of Jacobi matrices for generating such quadratures. Then, we consider differential operators acting on coefficients of a Jacobi expansion. Recent results due to Townsend, Oliver and Vasil enable the construction of sparse differential operators on the standard triangle, interoperating within the Jacobi polynomial family parameter space, and leading to low-storage, banded discrete differential operators. We generalize these results to the tetrahedra (laying the path for generalizing to $d > 3$), and use the resulting differential operators to construct the Eikonal operator on the simplex. Ultimately, we must solve a non-linear PDE, and working in coefficient space implies that our solution variables must be thought of as functions. Hence, we reformulate the Eikonal problem in terms of an infinite-dimensional L_2 minimization problem, which can be discretized with the aforementioned numerical quadrature. As with the README, this is more of a blog, but it's much easier to typeset math here.

Key words. multivariate orthogonal polynomials, spectral methods, partial differential equations, Eikonal equations, simplex

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1. Solutions to the Eikonal Equation. Haven't really looked into the viscosity perspective, but..I think I converged on the same idea for solving the PDE. Consider the standard simplex

$$\mathcal{T}^d = \{\mathbf{x} = (x_1, \dots, x_d) \in \mathbb{R}^d \mid x_i \geq 0, 1 - |\mathbf{x}|_1 \geq 0\}$$

The *minimal* time $u(x)$ to reach the boundary $\partial\mathcal{T}^d$ from a point $x \in \mathcal{T}^d$ is computable through solving the Eikonal equation on $\mathcal{T}^d$:

$$(1.1) \quad \begin{cases} \sum_{i=1}^d (\partial_{x_i} u)^2 = \left(\frac{1}{f}\right)^2, & x \in \mathcal{T}^d, \\ u = q, & x \in \partial\mathcal{T}^d, \end{cases}$$

where $f : \overline{\mathcal{T}^d} \rightarrow \mathbb{R}_*^+$ is the speed in the medium, $q : \partial\mathcal{T}^d \rightarrow \mathbb{R}^+$ is the time to leave $\mathcal{T}^d$ from a point on $\partial\mathcal{T}^d$ ($q \equiv 0$ if no exit cost). Splitting (1.1) into homogeneous and particular problems lends itself well to numerical approximation with optimization libraries. As such, we can solve the problem in two steps. First, solve for u_1 in

$$(1.2) \quad \begin{cases} \sum_{i=1}^d (\partial_{x_i} u_1)^2 = \left(\frac{1}{f}\right)^2, & x \in \mathcal{T}^d \end{cases}$$

with no boundary constraints. Then, solve for u_2 in

$$(1.3) \quad \begin{cases} \sum_{i=1}^d (\partial_{x_i} u_2)^2 = 0, & x \in \mathcal{T}^d \\ u_2 = q - u_1, & x \in \partial\mathcal{T}^d, \end{cases}$$

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with trivial RHS, and non-trivial boundary constraints. Noting that the value of u_1 on the boundary is annihilated in the combined solution

$$u = u_1 + u_2,$$

we conclude that u solves (1.1). **WRONG: I forgot to take account the fact of non-linearity in the PDO. What we really want is**

$$\begin{aligned} \left(\frac{1}{f}\right)^2 &= \sum_{i=1}^d (\partial_{x_i} u)^2 = \sum_{i=1}^d (\partial_{x_i} (u_1 + u_2))^2 = \sum_{i=1}^d (\partial_{x_i} u_1 + \partial_{x_i} u_2)^2 \\ &= \sum_{i=1}^d \partial_{x_i} u_1^2 + \partial_{x_i} u_2^2 + 2(\partial_{x_i} u_1)(\partial_{x_i} u_2) \end{aligned}$$

I missed those cross terms, which is one of the reasons I'm getting incorrect results as of now. We don't need to split the problem as such, so I will return to the original way of relying on the COBYLA algorithm for the fully constrained problem (1.1).

We seek to represent u under an optimal, finite, weighted L_2 expansion, wherein

$$u(\mathbf{x}) \approx \sum_{j=1}^n u_j P_j(\mathbf{x}) = \mathbf{u} \cdot \mathbf{P}(\mathbf{x}),$$

and $\mathbf{u}$ is the vector in $\mathbb{R}^n$ of coefficients of u under the elements of the basis $\mathbf{P}(\mathbf{x}) : \mathbb{R}^d \rightarrow \mathbb{R}^n$. Here $\approx$ must be taken in the sense of weighted L_2 convergence of the series to u as $n \rightarrow \infty$.

Let us specialize for $d = 2$. By P_k^n , we denote the k^{th} Jacobi polynomial on the triangle of degree n with parameters (a, b, c) . When the parameters must be specified, we will refer to the same polynomial by $P_{k,n}^{(a,b,c)}$. The weight function is

$$\begin{aligned} W^{(a,b,c)}(x, y) &= x^{a-\frac{1}{2}} y^{b-\frac{1}{2}} (1-x-y)^{c-\frac{1}{2}}, \\ w_{(a,b,c)} &= \left(\int_{\mathcal{T}_{\text{ref}}^2} W(x, y) dx dy \right)^{-1} = \frac{\Gamma(|\kappa| + \frac{3}{2})}{\Gamma(a + \frac{1}{2}) \Gamma(b + \frac{1}{2}) \Gamma(c + \frac{1}{2})} \\ w^{(a,b,c)}(\mathbf{x}) &= w_{(a,b,c)} W^{(a,b,c)}(\mathbf{x}) \end{aligned}$$

with $a, b, c > -1/2$ the Jacobi parameters, and $|\kappa| = a + b + c$. The orthonormal polynomials are given in terms of the 1D Jacobi polynomials by

$$\begin{aligned} P_k^n(x, y) &= (h_{k,n})^{-1} P_{n-k}^{2k+b+c, a-1/2} (2x-1) (1-x)^k P_k^{c-1/2, b-1/2} \left(\frac{2y}{1-x} - 1 \right), \\ (h_{k,n})^2 &= \frac{w_{a,b,c}}{(2n + |\kappa| + \frac{1}{2})(2k + b + c)} \\ &\quad \times \frac{\Gamma(n + k + b + c + 1) \Gamma(n - k + a + \frac{1}{2}) \Gamma(k + b + \frac{1}{2}) \Gamma(k + c + \frac{1}{2})}{(n - k)! k! \Gamma(n + k + |\kappa| + \frac{1}{2}) \Gamma(k + b + c)}, \end{aligned}$$

and they satisfy a mutual orthonormality condition under the weighted inner product:

$$\langle P_{j,m}^{(a,b,c)}, P_{k,n}^{(a,b,c)} \rangle_{w^{(a,b,c)}} = \delta_{jk} \delta_{mn}.$$

That is to say, a given basis polynomial is not only orthonormal to those with different total degree, but also to those others within the same-degree homogeneous subspace

with that degree. Notice, P_k^n , $0 \leq k \leq n$ forms a basis for the space of homogeneous polynomials of degree n . By considering ordered collections of these subspaces (where ordering is in terms of the degree n of the space), instead of individual polynomials, we can more elegantly reformulate the Eikonal problem into a weak-type form. Let

$$\mathbb{P}_n(x_1, x_2) = (P_0^n(x_1, x_2), \dots, P_n^n(x_1, x_2))^T,$$

and let

$$\mathbf{P} = (\mathbb{P}_0^T, \mathbb{P}_1^T, \dots, \mathbb{P}_{n-1}^T)^T.$$

54 The above notation amounts to a vectorization of the basis for the general space of
55 polynomials in two variables Π_{n-1}^2 defined of $\mathcal{T}^2$.

56 Now, consider the $(n-1)$ order Jacobi polynomial expansion of the solution u
57 under the parameter family (a, b, c) . For $\mathbf{x} \in \mathcal{T}^d$, and $N = \dim \Pi_{n-1}^2$ coefficients
58 $\mathbf{u} \in \mathbb{R}^N$, we can write it as

$$59 \quad (1.7) \quad u(\mathbf{x}) = \mathbf{P}^{a,b,c}(\mathbf{x})^T \mathbf{u}.$$

60 There exists sparse discrete operators which map $\mathbf{u}$ to the coefficients of the derivative
61 of u under a modified basis, as well as those which promote the basis parameters
62 without taking derivatives:

$$63 \quad (1.8) \quad \begin{aligned} \partial_{x_1} u(\mathbf{x}) &= \mathbf{P}^{(a+1,b+1,c+1)}(\mathbf{x})^T \mathcal{K}_{(a+1,b,c+1)}^{(a+1,b+1,c+1)} \mathcal{D}_{x_1} \mathbf{u} \\ \partial_{x_2} u(\mathbf{y}) &= \mathbf{P}^{(a+1,b+1,c+1)}(\mathbf{x})^T \mathcal{K}_{(a,b+1,c+1)}^{(a+1,b+1,c+1)} \mathcal{D}_{x_2} \mathbf{u} \end{aligned}$$

64 Inserting this into the fully constrained Eikonal equation gives

$$65 \quad (1.9) \quad \left(\mathbf{P}^{(a+1,b+1,c+1)}(\mathbf{x})^T \mathcal{K}_{(a,b,c)}^{(a+1,b+1,c+1)} \mathbf{c}_{\frac{1}{f}} \right)^2 =$$

$$66 \quad \left(\mathbf{P}^{(a+1,b+1,c+1)}(\mathbf{x})^T \mathcal{K}_{(a+1,b,c+1)}^{(a+1,b+1,c+1)} \mathcal{D}_{x_1} \mathbf{u} \right)^2 +$$

$$67 \quad \left(\mathbf{P}^{(a+1,b+1,c+1)}(\mathbf{x})^T \mathcal{K}_{(a,b+1,c+1)}^{(a+1,b+1,c+1)} \mathcal{D}_{x_2} \mathbf{u} \right)^2 =$$

$$68 \quad (1.10) \quad \mathbf{P}^{(a+1,b+1,c+1)}(\mathbf{x})^T \left(\tilde{\mathcal{D}}_{x_1} \mathbf{u} \mathbf{u}^T \tilde{\mathcal{D}}_{x_1}^T + \tilde{\mathcal{D}}_{x_2} \mathbf{u} \mathbf{u}^T \tilde{\mathcal{D}}_{x_2}^T \right) \mathbf{P}^{(a+1,b,c+1)}(\mathbf{x}),$$

where

$$\tilde{\mathcal{D}}_{x_1} = \mathcal{K}_{(a+1,b,c+1)}^{(a+1,b+1,c+1)} \mathcal{D}_{x_1}, \quad \tilde{\mathcal{D}}_{x_2} = \mathcal{K}_{(a,b+1,c+1)}^{(a+1,b+1,c+1)} \mathcal{D}_{x_2}$$

69 and, the coefficients of $(1/f)$ are represented as $\mathbf{c}_{\frac{1}{f}}$ but promoted to the $(a+1, b+$
70 $1, c+1)$ Jacobi basis. We let $\tilde{\cdot}$ indicate promotion *to* $(a+1, b+1, c+1)$ from any other
71 parameter set. The above equality holds for any point $\mathbf{x} \in \mathcal{T}^2$. So, for a solution $\mathbf{u}$,
72 the coefficient operation must be such that:

$$73 \quad (1.11) \quad \left(\tilde{\mathcal{D}}_{x_1} \mathbf{u} \mathbf{u}^T \tilde{\mathcal{D}}_{x_1}^T + \tilde{\mathcal{D}}_{x_2} \mathbf{u} \mathbf{u}^T \tilde{\mathcal{D}}_{x_2}^T \right) = \tilde{\mathbf{c}}_{\frac{1}{f}} \tilde{\mathbf{c}}_{\frac{1}{f}}^T.$$

74 Now, we seek a coefficient solution $\mathbf{u}$ which represents the solution *globally* in $\mathcal{T}^2$.
75 However, polynomials are differentiable, and some of the problems we are interested
76 in yield non-differentiable solutions. This is true even in the most basic case of $f \equiv 1$,
77 in which u is the signed distance function from $\mathbf{x}$ to $\partial \mathcal{T}^2$, depicted below:

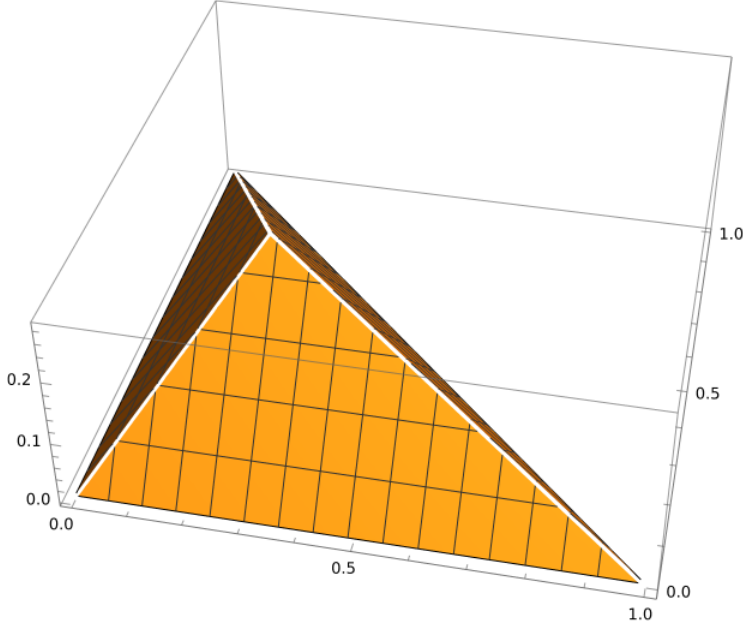


FIG. 1. A simple calculation shows that the minimum distance from $\mathbf{x} \in \mathcal{T}^2$ to the boundary $\partial\mathcal{T}^2$ is given by the minimum of $\left\{x_1, x_2, \left\| \mathbf{x} - \begin{bmatrix} (x_1 - x_2 + 1)/2 \\ (x_2 - x_1 + 1)/2 \end{bmatrix} \right\|_2 \right\}$

78 We cannot hope to recover a solution $\mathbf{u}$ which satisfies (1.11) to machine precision,
 79 as that would imply the global solution has zero residual *pointwise* everywhere. Hence,
 80 we must return to (1.9), and consider solving it weakly, in the sense of:

$$\begin{aligned}
 (1.12) \quad & \int_{\mathcal{T}^2} w^{(a+1,b+1,c+1)}(\mathbf{x}) \mathbf{P}^{(a+1,b+1,c+1)}(\mathbf{x})^T \tilde{\mathbf{c}}_{\frac{1}{f}}^T \tilde{\mathbf{c}}_{\frac{1}{f}} \mathbf{P}^{(a+1,b+1,c+1)}(\mathbf{x}) d\mathbf{x} \approx \|1/f\|_{w^{(a+1,b+1,c+1)}}^2 \\
 81 \quad & \approx \int_{\mathcal{T}^2} w^{(a+1,b+1,c+1)}(\mathbf{x}) \mathbf{P}^{(a+1,b+1,c+1)}(\mathbf{x})^T (\tilde{\mathcal{D}}_{x_1} \mathbf{u} \mathbf{u}^T \tilde{\mathcal{D}}_{x_1}^T + \tilde{\mathcal{D}}_{x_2} \mathbf{u} \mathbf{u}^T \tilde{\mathcal{D}}_{x_2}^T) \mathbf{P}^{(a+1,b+1,c+1)}(\mathbf{x}) d\mathbf{x} \\
 & \approx \|\partial_x u\|_{w^{(a+1,b+1,c+1)}}^2 + \|\partial_y u\|_{w^{(a+1,b+1,c+1)}}^2. \quad \blacksquare
 \end{aligned}$$

82 Here the norm is viewed as that of a function in $L^2(\mathcal{T}^2, w^{(a+1,b+1,c+1)})$.

Parsing this equation out a bit further (condensing sub/super scripts), we have

$$\begin{aligned}
 & \int_{\mathcal{T}^2} \mathbf{P}(\mathbf{x})^T (\mathbf{u}_x \mathbf{u}_x^T + \mathbf{u}_y \mathbf{u}_y^T) w(\mathbf{x}) \mathbf{P}(\mathbf{x}) d\mathbf{x} \quad (\text{Only } x \text{ part handled below}) \\
 &= \int_{\mathcal{T}^2} \mathbf{P}(\mathbf{x})^T \begin{bmatrix} u_{x,1}^2 & u_{x,1}u_{x,2} & \cdots & u_{x,1}u_{x,N} \\ u_{x,2}u_{x,1} & u_{x,2}^2 & \cdots & u_{x,2}u_{x,N} \\ \vdots & \ddots & \ddots & \vdots \\ u_{x,N}u_{x,1} & u_{x,N}u_{x,2} & \cdots & u_{x,N}^2 \end{bmatrix} \mathbf{P}(\mathbf{x}) w(\mathbf{x}) d\mathbf{x} \\
 &= \int_{\mathcal{T}^2} \begin{bmatrix} P_1(\mathbf{x}) & P_2(\mathbf{x}) & \cdots & P_N(\mathbf{x}) \end{bmatrix} \begin{bmatrix} u_{x,1}^2 & u_{x,1}u_{x,2} & \cdots & u_{x,1}u_{x,N} \\ u_{x,2}u_{x,1} & u_{x,2}^2 & \cdots & u_{x,2}u_{x,N} \\ \vdots & \ddots & \ddots & \vdots \\ u_{x,N}u_{x,1} & u_{x,N}u_{x,2} & \cdots & u_{x,N}^2 \end{bmatrix} \begin{bmatrix} P_1(\mathbf{x}) \\ P_2(\mathbf{x}) \\ \vdots \\ P_N(\mathbf{x}) \end{bmatrix} w(\mathbf{x}) d\mathbf{x} \\
 &= \int_{\mathcal{T}^2} \begin{bmatrix} P_1(\mathbf{x}) & P_2(\mathbf{x}) & \cdots & P_N(\mathbf{x}) \end{bmatrix} \begin{bmatrix} u_{x,1} \sum_{j=1}^N u_{x,j} P_j(\mathbf{x}) \\ u_{x,2} \sum_{j=1}^N u_{x,j} P_j(\mathbf{x}) \\ \vdots \\ u_{x,N} \sum_{j=1}^N u_{x,j} P_j(\mathbf{x}) \end{bmatrix} w(\mathbf{x}) d\mathbf{x} \\
 &= \int_{\mathcal{T}^2} w(\mathbf{x}) \sum_{k=1}^N P_k(\mathbf{x}) u_{x,k} \sum_{j=1}^N u_{x,j} P_j(\mathbf{x}) d\mathbf{x} = \int_{\mathcal{T}^2} w(\mathbf{x}) \sum_{k=1}^N \sum_{j=1}^N u_{x,k} u_{x,j} P_k(\mathbf{x}) P_j(\mathbf{x}) d\mathbf{x} \\
 &= \int_{\mathcal{T}^2} \sum_{k=1}^N u_{x,k}^2 P_k^2(\mathbf{x}) w(\mathbf{x}) d\mathbf{x} = \sum_{k=1}^N u_{x,k}^2 \int_{\mathcal{T}^2} P_k^2(\mathbf{x}) dw(\mathbf{x}) \quad (\text{now we add back in } y \text{ part}) \\
 &= \sum_{k=1}^N u_{x,k}^2 + \sum_{k=1}^N u_{y,k}^2 \\
 &= \mathbf{u}^T (\tilde{\mathcal{D}}_x^T \tilde{\mathcal{D}}_x + \tilde{\mathcal{D}}_y^T \tilde{\mathcal{D}}_y) \mathbf{u} = \tilde{\mathbf{c}}_{\frac{1}{f}}^T \tilde{\mathbf{c}}_{\frac{1}{f}}
 \end{aligned}$$

where we've used the mutual orthonormality relations, which persist even within a subspace of homogeneous polynomials of degree m , to eliminate all off-diagonal terms in the sum. Note, not all orthonormal bases are created equal, and we normally would have to retain many outer product terms in the sum because orthonormality between vectors of the same subspace is not a given.

Hence, we can frame the discrete Eikonal solver in terms of a quadratic program with equality constraints:

$$\begin{aligned}
 (1.13) \quad & \min_{\mathbf{u} \in \mathbb{R}^n} \frac{1}{2} \mathbf{u}^T G \mathbf{u} + \mathbf{u}^T c \\
 & \text{subject to } A \mathbf{u} = b,
 \end{aligned}$$

where $G = \tilde{\mathcal{D}}_x^T \tilde{\mathcal{D}}_x + \tilde{\mathcal{D}}_y^T \tilde{\mathcal{D}}_y \in \mathbb{R}^{N \times N}$ is symmetric, and we need to derive the form of A for enforcing conditions on the boundary. If u is the time to exit, and f is the speed in the medium, then there must be a way to write f in terms of u , so that we can cast the problem completely as a QP. However, this is a non-linear relation, so I'm not sure yet how to do this with linear operations. Perhaps we proceed weakly as

$$(1.14) \quad \|fu\|_{L^2(\mathcal{T}^2, w^{(a+1, b+1, c+1)})} = \alpha$$

and use Parseval's theorem? Then, $\|fu\| = \dots$ I'm tired and need to think/sleep/think tomorrow. But, I'm hoping that we can let $c \equiv 0$, and incorporate the speed into the constraint relation.

2. A variational approach. Let's try framing the problem from the perspective of infinite dimensional optimization. Thus far, my efforts have fallen under the discretize-then-optimize approach for solving non-linear PDEs. So, let's flip that and consider the optimize-then-discretize approach.

We want to solve

$$(2.1) \quad \begin{aligned} \min_{u \in L_2(\mathcal{T}^2, w^{(a,b,c)})} \mathcal{L}(u), \\ \text{s.t. } u = q \text{ on } \partial\mathcal{T}^2, \end{aligned}$$

where $\mathcal{L}$ is defined by

$$(2.2) \quad \mathcal{L}(u) := \int_{\mathcal{T}^2} (|\nabla_{\mathbf{x}} u|^2 - 1) d\mathbf{x}$$

The steps here are to derive the first variation of $\mathcal{L}$ as

$$\delta\mathcal{L}(u, \hat{u}) = \partial_\varepsilon \mathcal{L}(u + \varepsilon \hat{u})|_{\varepsilon=0} = 0,$$

where $\hat{u} \in \mathcal{H}_0^1(\mathcal{T}^2)$. This gives the necessary condition a minimizer of $\mathcal{L}$ must satisfy. Then, compute the second variation by considering another $\tilde{u} \in \mathcal{H}_0^1$, and deriving the newton step through

$$\delta^2\mathcal{L}(u, \hat{u}, \tilde{u}) = \partial_\varepsilon \delta\mathcal{L}(u + \varepsilon \tilde{u}, \hat{u})$$

and find the step direction by solving

$$\delta^2\mathcal{L}(u^k)(\hat{u}, \tilde{u}) = -\delta\mathcal{L}(u^k)(\hat{u}), \quad \forall \hat{u} \in \mathcal{H}_0^1$$

By choosing $u^0 = q$ on $\partial\mathcal{T}^2$, and setting $u^{k+1} = u^k + \tilde{u}$, we have an infinite-dimensional newton stepper for solving the Eikonal problem. I'm not sure I like where this goes in terms of complexity (mental and computational), so will stick to the DTO approach for now. There is also probably a correct energy functional which gives the Eikonal problem, which is likely not the one I started with.